Meeting 4 - 31/07/2026

Mainly discussed abt architecture

Key Discussion Points

Energy Module Architecture

- **Centralised:** A single energy module manages energy calculations for all satellites. Simple to manage, but creates a central dependency and may not scale as well.
- **Localised:** Each satellite/container manages its own energy state and calculations. More distributed and independent, but can duplicate logic across containers.
- **Sidecar:** A separate energy container runs alongside each satellite container and handles its energy model. Keeps energy logic isolated, but adds additional containers and resource overhead.

Research pros/cons of each, and remember, the current approach should be simple. We might change the architecture later!

Core Design Goal

The energy extension should be:

- **Generic**
- **Extensible**
- **Simple**
- **Easy for users to configure**
- Designed to support **multiple energy models (generation/consumption)** in the future

The goal is **not to build the most complete model immediately**. Start with the simplest useful implementation and design the architecture so more sophisticated models can be added later.

Energy Model Design

- Implement the **simplest energy model first**.
- The core architecture should support adding alternative energy models later.
- New models should be addable without significantly modifying existing code.
- The implementation should include a clear example showing users **where and how to change or add a model**.
- Avoid unnecessary complexity in the initial version.

Engineering Principles

- Apply the **Open/Closed Principle**:
  - Open for extension
  - Closed for unnecessary modification
- Think from the **user's perspective**, not only the developer's perspective.
- Consider realistic scenarios for how someone would actually configure and run the emulator.
- Keep configuration and setup as simple as possible.
- Use short, useful comments inside the code. (Current OpenSN code has no comments)
- **<u>If an existing OpenSN design creates unnecessary limitations, consider improving it rather than blindly preserving it.</u>**

Architecture

Propose a design plan next meeting!!! Will discuss SWE best practices

Consider:

- What functionality belongs in the **core**?
- How should different energy models communicate with the rest of OpenSN?
- What information needs to be stored in the database
